Bitcoin Transaction Network Analysis

Fraud Risk Compliance Intelligence Report

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Name: Opeyemi Mercy Lawan
Email: Messilawan2008@gmail.com
Tools: Google Colab, Microsoft Word, Microsoft PowerPoint, Github

Executive Summary

This report presents findings from a Graph Neural Network (GNN) fraud detection analysis applied to the complete Elliptic Bitcoin transaction dataset. The dataset comprises **203,769 Bitcoin transactions** and **234,355 directed payment flows** spanning 49 two-week periods of Bitcoin network activity.

A **GraphSAGE** model is an inductive graph neural network capable of learning from both transaction features and network structure was trained on 29,894 labelled transactions (time steps 1–34) and evaluated on 16,670 held-out transactions (time steps 35–49). The model then generated fraud probability scores for **157,205** previously unclassified wallets that constitute 77% of the dataset.

The central finding of this analysis is that the original dataset's 4,545 confirmed illicit wallets significantly understate the true at-risk population. The GNN identifies **88,234 additional high-risk wallets**; a **1,971% increase** in detectable suspicious activity. These wallets exhibit the same structural patterns, neighborhood profiles, and temporal clustering as confirmed illicit nodes, but had previously escaped classification.

Key Findings at a Glance:

- AUC-ROC of 0.957: the model correctly ranks a fraudulent wallet above a legitimate one in 95.7% of cases
- Recall of 0.910: the model catches 9 out of every 10 confirmed illicit wallets in the test period
- 88,234 previously unknown wallets identified as high-risk
- 6 statistically anomalous burst windows identified at time steps 9, 13, 20, 28, 29, and 32
- Illicit activity propagates to 35.57% of exposed downstream wallets on average - peaking at 84% at time, step 29
- Fraud is structurally organized: not randomly consistent with coordinated money laundering operations

Model Performance

The GraphSAGE model was trained using a **temporal train/test split**, training on the first 34-time steps and testing on the final 15. This approach mirrors real-world deployment conditions, where a fraud model must generalize about future transactions it has never seen during training. The following metrics were recorded on the held-out test set:

AUC-ROC	Recall (Illicit)	F1 Score	Avg Precision
0.957	0.910	0.424	0.838

Standard classification accuracy is not reported here as it is a misleading metric on severely imbalanced data. A model that labelled every transaction as licit would achieve 90% accuracy while detecting zero fraud. Instead, **recall and average precision** are the primary evaluation metrics here, as missing a fraudster carries far greater regulatory and financial cost than investigating a false positive.

Confusion Matrix Summary (test set): Of 1,083 confirmed illicit transactions in the test period, the model correctly identified 982 (91%). Only 101 illicit transactions were missed. Among 15,587 licit transactions, 13,021 were correctly identified, with 2,566 flagged as false positives which is a manageable false alarm rate given the stakes.

Detection Uplift for Full Network

When applied to all 157,205 unclassified wallets, the GNN substantially expands the identifiable at-risk population:

Category	Node Count	% of Network
Confirmed Illicit (labelled)	4,545	2.2%
GNN-Predicted High-Risk (unknown)	88,234	43.3% of unknowns
Total At-Risk Pool	92,779	1,971% increase
Confirmed Licit	42,019	20.6%
Unresolved Unknown	69,971	34.3% of unknowns

The 69,971 wallets with GNN illicit probability below 0.5 are classified as predicted-licit but remain unverified by ground-truth labelling. These should not be treated as confirmed clean until cross-referenced against external intelligence sources (see Recommendation 3).

Five Highest-Risk Wallets

The following wallets carry the highest GNN-assigned fraud probabilities in the network. All five were previously unclassified, they carried no label in the original Elliptic dataset. Risk rationales are derived from the model's feature attribution, network position analysis, and temporal context

Rank	Transaction ID	Status	GNN Probability	Time Step	Risk Rationale
#1	97,869,259	Unknown	99.994%	40	Active in burst window t=40. High out-degree fan-out pattern.

					Neighbourhood dominated by predicted illicit predecessors.
#2	378,429,614	Unknown	99.991%	16	Bridge node with high betweenness centrality. Sits on many shortest paths between other high-risk nodes and structural hallmark of a money-laundering intermediary.
#3	210,915,044	Unknown	99.991%	20	Active at burst step t=20 (z=2.1σ). Predecessor illicit ratio 0.87(87%) of nodes that sent funds to this wallet are themselves flagged as illicit.
#4	99,916,860	Unknown	99.990%	21	Consecutive time step activity (t=20, t=21) with near-identical risk profile to #3 is consistent with a chain of disposable layering wallets.
#5	97,810,198	Unknown	99.990%	40	Second high-risk wallet active at burst step t=40. Co-active with #1 in the same time window suggests coordinated activity.

Note: 'Unknown' status means these wallets were not labelled in the original dataset. It does not mean they have been verified as clean. GNN fraud probabilities above 99.9% represent near-certain predictions based on structural, temporal, and neighbourhood evidence.

Most Suspicious Cluster - Burst Window Analysis

Statistical anomaly detection was applied to the illicit rate across all 49 time steps. Time steps where the illicit rate exceeds **1.5 standard deviations above the mean** (z-score > 1.5) were flagged as burst windows. Six such windows were identified:

Time Step	Illicit Rate	Z-Score	Illicit Count	Significance
13	35.2%	2.57σ	High	Highest illicit rate in entire dataset. 1 in 3 labelled nodes is illicit
20	29.1%	2.09σ	High	Second highest rate. 260 illicit nodes active in single window
9	31.5%	1.93σ	Elevated	Early burst, signals the onset of organised fraud campaign
28	30.2%	1.79σ	Elevated	Peaks at 84% propagation rate in t=29. The highest spread in dataset
29	28.9%	1.87σ	Elevated	84.11% of exposed nodes became illicit. This is the strongest propagation in entire network
32	25.8%	1.52σ	Borderline	342 illicit nodes, the highest absolute count across all time steps

Interpretation: The non-random distribution of burst windows occurring roughly every 5–10 time steps is inconsistent with opportunistic individual fraud. The pattern is more consistent with organised, coordinated fraud campaigns that operate in cycles of activity and dormancy. This is a finding of material significance for both the investigations and compliance monitoring functions.

Of particular note is the **step 13 window**, where the illicit rate reached 35.2% meaning more than 1 in 3 labelled wallets active in that period were confirmed fraudulent. This is 3.1 times the network-wide mean illicit rate of 11.35% among labelled nodes.

Recommendations

A key question for compliance is not just **who** is fraudulent, but **how** fraud spreads through the network over time. For each time step T, the analysis identified all nodes that received funds from confirmed illicit sources at T, then measured what proportion of those exposed nodes were themselves illicit at T+1. The actions below are sequenced by urgency. Both have a named owner and a clear deadline.

Recommendation	Action Required	Owner & Deadline
1. Immediate SAR Assessment on Top-5 GNN-Flagged Wallets	Wallets 97,869,259 · 378,429,614 · 210,915,044 · 99,916,860 · 97,810,198 each carry a GNN-assigned fraud probability above 99.99%. This level of model confidence, derived from the wallet's own transaction features, its position in the network, and the behaviour	Head of Investigations Deadline: 10 working days

	of every wallet it has transacted with is the strongest possible signal the system can produce. The Investigations team should obtain full transaction histories for each wallet, identify all counterparties, and assess whether a Suspicious Activity Report is warranted under current AML obligations. If no legitimate business purpose can be established within 10 working days, a SAR should be filed and account restriction considered. Given that three of the five wallets were active statistically confirmed burst windows (time steps 20, 28, and 40) the counterparty chains from those periods should be prioritised for chain-of-funds tracing.	
2. Enhanced Monitoring of All Burst-Window Transactions	Six time steps 9, 13, 20, 28, 29, and 32 — have been identified as statistically anomalous using z-score analysis, each recording an illicit rate more than 1.5 standard deviations above the network-wide mean. Time step 13 is the most extreme, with a confirmed illicit rate of 35.2% meaning more than one in three labelled wallets active in that window were fraudulent. The Compliance Monitoring team should apply Level 2 enhanced due diligence to all wallets active in these windows. Particular attention should be given to the step 28–29 cluster, where the propagation analysis recorded the highest fraud spread in the entire dataset 84.11% of wallets that received funds from illicit sources at step 28 were themselves illicit at step 29. This is consistent with a rapid layering operation in which funds were passed through a chain of disposable wallets over a very short period. All downstream wallets receiving funds from step 28–29 activity should be flagged for chain-of-funds tracing. Estimated scope: 12,000–15,000 wallets.	Head of Compliance Monitoring Deadline: 20 working days

3. External Classification of 157,205 Unclassified Wallets	77% of the network currently carries no ground-truth label. The GNN has assigned fraud probabilities to all 157,205 unknown wallets, identifying 88,234 as high-risk and 69,971 as predicted-licit but these remain model predictions, not confirmed classifications. The Data & Analytics team should commission a cross-reference of all unknown wallets against external blockchain intelligence providers such as Chainalysis, Elliptic, or TRM Labs. Priority should be given to the 12,436 unknown wallets that were active during the six identified burst windows, as these carry the highest contextual risk regardless of their individual GNN scores. Results should be fed back into the GNN model as new ground-truth labels to enable continuous model improvement. Until the classification gap is resolved, all Unknown wallets with a GNN illicit probability above 0.7, approximately 34,000 wallets, should be treated as elevated-risk for transaction monitoring purposes.	Head of Data & Analytics Deadline: 30 working days

Data Limitation

Four limitations were observed from the dataset used to accomplish this project.

Limitations Detected	Explanation
1. 77% of the Network is Unlabelled	The most significant constraint on this analysis is that 157,205 of the 203,769 wallets in the dataset carry no ground-truth label. The GNN's predictions on these wallets are probabilistic estimates derived from structural patterns, temporal context, and neighbourhood behaviour — they are not confirmed fraud determinations. The model achieves a recall of 91% on labelled test data, which means approximately 9% of genuinely illicit wallets may be assigned a low fraud probability and escape detection. In absolute terms, with 4,545

	confirmed illicit wallets in the dataset, this implies the model may miss in the region of 400–500 additional illicit wallets whose behaviour is sufficiently atypical to evade detection. The 92,779-node at-risk pool identified by this analysis should therefore be treated as a lower bound on total exposure, not a ceiling. This limitation will not be fully resolved until Recommendation 3 is implemented.
2. Temporal Scope of the Dataset	The Elliptic dataset covers Bitcoin network activity from 2013 to 2014. Bitcoin transaction patterns, wallet behaviour, mixer and tumbler usage, and fraud methodologies have evolved substantially in the decade since this data was collected. The emergence of privacy coins, cross-chain bridges, decentralised exchanges, and more sophisticated layering techniques means that the structural signatures of illicit activity in contemporary networks may differ meaningfully from those captured here. The model's performance metrics — AUC 0.957, Recall 0.910 — reflect accuracy on 2013–2014 data evaluated on a held-out subset of that same period. These figures should not be assumed to transfer directly to current Bitcoin network activity without retraining on contemporary labelled data. Regulatory and compliance decisions informed by this analysis should account for this temporal gap.
3. Feature Opacity	The 166 node features provided by the Elliptic dataset are anonymised. Their precise financial interpretations — what specific transaction characteristics each feature encodes — are not disclosed by the dataset providers. This limits the ability to provide fully human-interpretable rationales for individual node risk scores. While structural evidence (network position, neighbourhood composition, temporal context) remains transparent and explicable, the contribution of the original 166 features to any given prediction cannot be fully decoded into plain-language financial terms. This is relevant for regulatory reporting purposes, where explainability of model decisions may be required under applicable AML frameworks. A feature attribution analysis using

	SHAP values is recommended as a follow-on study to address this gap.
4. Early Warning Predictive Accuracy	The autoregressive early warning model — which attempts to predict the next time step's illicit rate from the previous three time steps — achieves a mean absolute error of 0.058 against a mean illicit rate of 0.120, representing a 48.6% error relative to the mean. While the model captures the general trend of illicit activity over time, it does not reliably anticipate the sharp burst spikes that represent the highest-risk periods. This suggests that burst events are triggered by external coordination factors — such as darknet market activity, exchange delistings, or law enforcement actions — that are not captured in the transaction time series alone. A more robust early warning system would require the integration of external signals beyond the scope of this dataset.